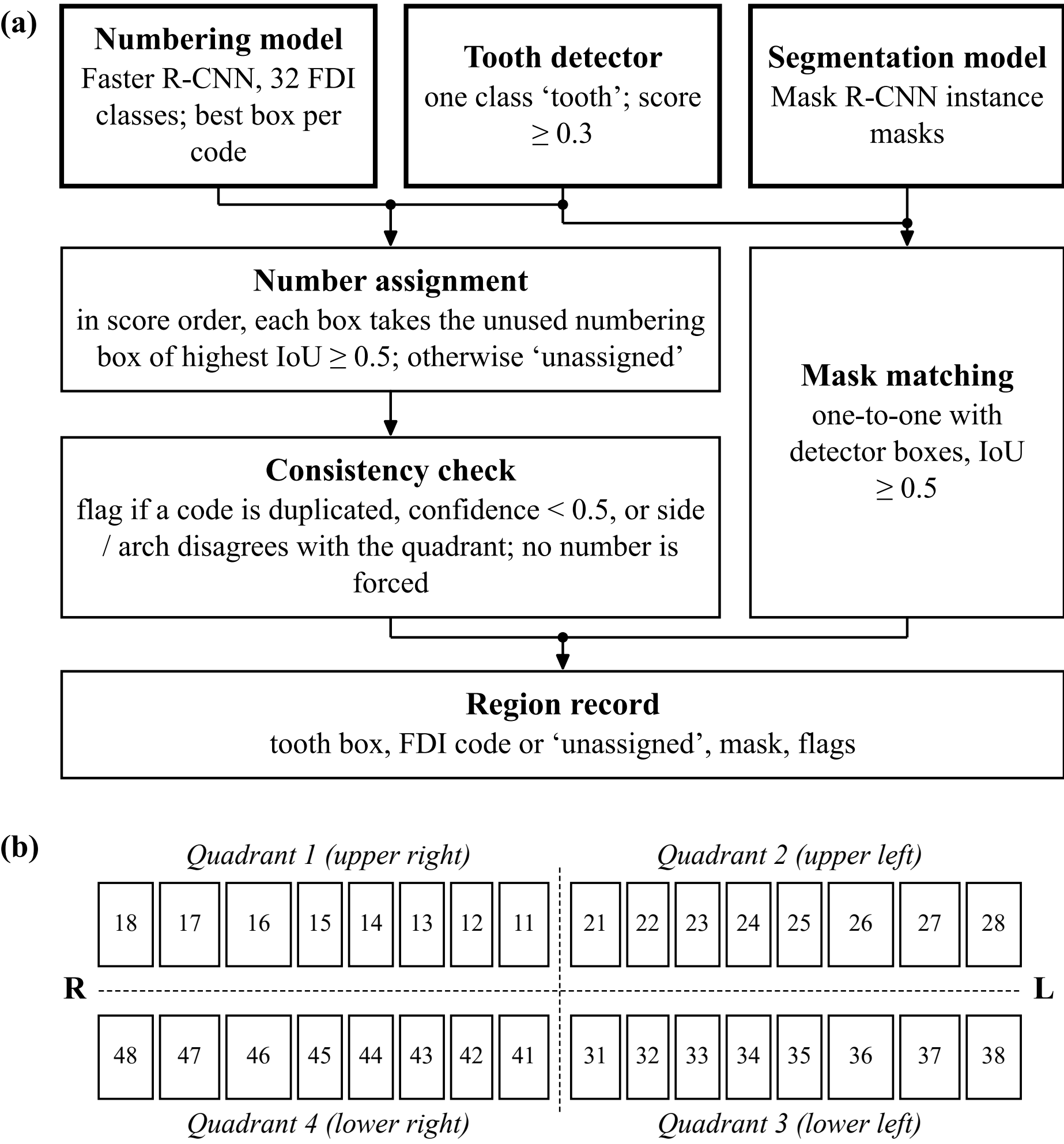

(a)
Numbering model
Faster R-CNN, 32 FDI classes; best box per code
Tooth detector
one class 'tooth'; score ≥ 0.3
Segmentation model
Mask R-CNN instance masks
Number assignment
in score order, each box takes the unused numbering box of highest IoU ≥ 0.5; otherwise 'unassigned'
Mask matching
one-to-one with detector boxes, IoU ≥ 0.5
Consistency check
flag if a code is duplicated, confidence < 0.5, or side / arch disagrees with the quadrant; no number is forced
Region record
tooth box, FDI code or 'unassigned', mask, flags
(b)
Quadrant 1 (upper right)
Quadrant 2 (upper left)
18 17 16 15 14 13 12 11
21 22 23 24 25 26 27 28
R
L
48 47 46 45 44 43 42 41
31 32 33 34 35 36 37 38
Quadrant 4 (lower right)
Quadrant 3 (lower left)